

ALEXEY GALLEGO

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SUMMARY

Throughout my career, I have repeatedly been trusted with problems I had never solved before and technologies I had never used before. Whether developing industrial automation systems, leading ERP initiatives, building business software, mentoring engineers, or adopting emerging AI workflows, I have built a reputation for learning quickly, taking ownership, and delivering results. My passion is helping organizations grow by turning complex operational challenges into scalable systems, products, and processes.

TECHNICAL SKILLS

Languages & Frameworks: JavaScript, TypeScript, Node.js, React, Next.js, HTML, CSS, Tailwind CSS

AI: MCP, OpenAI APIs, Claude APIs, Prompt Engineering, RAG, Codex, Claude Code.

Databases: SQL Server, PostgreSQL, MySQL, MongoDB

Infrastructure & DevOps: Docker, Kubernetes, Linux, CI/CD, GitHub Actions, AWS, Nginx, Ubuntu Server, VMware

Methodologies: Agile/Scrum, Technical Leadership, Code Reviews, Mentorship.

EXPERIENCE

Systems & Software Engineer | **Encore Automation** | Auburn Hills, MI *Jul 2021 – Present*

- Directed the technical transformation of a company-wide ERP initiative, leading two engineers while defining architecture, security, infrastructure, development standards, and deployment strategy across engineering, purchasing, finance, receiving, and project management operations.
- Identified operational inefficiencies affecting more than 100 employees and architected a company-wide business operations platform from scratch (Node.js, React, SQL), replacing spreadsheet-driven processes with automated timesheets, expense management, scheduling, approval workflows, and reporting that improved visibility and accountability across the organization.
- Increased timesheet compliance from approximately 70% to 95% and expense submission compliance from approximately 50% to 90% by redesigning user workflows, implementing automated notifications, and creating manager dashboards that eliminated manual administrative follow-up.
- Led technical architecture and infrastructure decisions for internal business applications, maintaining 99%+ uptime through ownership of Ubuntu-based production environments, Dockerized deployments, CI/CD pipelines, and application monitoring.
- Strengthened company security posture by replacing legacy authentication approaches with modern JWT-based authentication, refresh token rotation, CSRF protection, secure password management, and role-based access controls across multiple business systems.
- Rebuilt the company's digital presence using Next.js, modernizing product marketing, improving mobile usability, increasing online visibility, and creating a stronger platform for customer engagement and lead generation.
- Introduced AI-assisted engineering workflows and intelligent receipt processing capabilities that reduced administrative review effort from approximately one day to one to two hours while accelerating software delivery and improving operational scalability.
- Mentored multiple engineering interns across software and automation projects, providing technical guidance, architecture reviews, and hands-on support while fostering engineering growth and development.

PROJECTS

AI Receipt Analysis System

- Designed and implemented an AI-powered expense processing pipeline using OpenAI Vision that analyzes receipt images, extracts structured line-item data, validates submissions, and automatically matches expenses to company records. Integrated as a core ERP module, reducing administrative review effort from approximately one business day to one to two hours while improving accuracy and scalability.

Industrial Process Validation System

- Developed a Python-based industrial quality monitoring platform that interfaces directly with FANUC robots, captures real-time process telemetry, automatically identifies defective components, and provides production visibility dashboards. Improved process reliability from approximately 80% to 99% while reducing manufacturing risk and improving quality assurance.

EDUCATION

B.S. Mechanical Engineering | Florida International University, Miami, FL | May 2021